

lab03_Goes

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1

```
library("readr")
```

Warning: pacote 'readr' foi compilado no R versão 4.5.3

```
airlines <- read_csv("airlines.csv")
```

Rows: 14 Columns: 2

-- Column specification -----

Delimiter: ","

chr (2): IATA_CODE, AIRLINE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
airports <- read_csv(  
  "airports.csv",  
  col_select = c(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE)  
)
```

Rows: 322 Columns: 5

-- Column specification -----

Delimiter: ","

chr (3): IATA_CODE, CITY, STATE

dbl (2): LATITUDE, LONGITUDE

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
library(dplyr)
```

Warning: pacote 'dplyr' foi compilado no R versão 4.5.3

Anexando pacote: 'dplyr'

Os seguintes objetos são mascarados por 'package:stats':

filter, lag

Os seguintes objetos são mascarados por 'package:base':

intersect, setdiff, setequal, union

```
library(stringr)
```

Warning: pacote 'stringr' foi compilado no R versão 4.5.3

```
processar_bloco <- function(chunk, pos) {  
  chunk %>%  
    filter(!str_starts(DESTINATION_AIRPORT, "1")) %>%  
    filter(!is.na(DESTINATION_AIRPORT), !is.na(ARRIVAL_DELAY)) %>%  
    group_by(DESTINATION_AIRPORT) %>%  
    summarise(  
      soma_atraso = sum(ARRIVAL_DELAY),  
      total_voos = n(),  
      .groups = "drop"  
    )  
}  
  
estatisticas_acumuladas <- read_csv_chunked(  
  "flights.csv",  
  callback = DataFrameCallback$new(processar_bloco),  
  chunk_size = 1000000,  
  col_types = cols_only(  
    DESTINATION_AIRPORT = col_character(),  
    ARRIVAL_DELAY = col_double()  
  )  
)
```

```

flights <- estadisticas_acumuladas %>%
  group_by(DESTINATION_AIRPORT) %>%
  summarise(
    atraso_medio = sum(soma_atraso) / sum(total_voos),
    .groups = "drop"
  )

```

2

```

setdiff(flights$DESTINATION_AIRPORT, airports$IATA_CODE)

```

```

character(0)

```

```

flights <- flights %>%
  left_join(
    airports %>% select(IATA_CODE, CITY, STATE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  )

```

3

```

aeropertos_por_estado <- airports %>%
  filter(!is.na(STATE)) %>%
  count(STATE, name = "num_aeropertos") %>%
  arrange(desc(num_aeropertos))

```

```

aeropertos_por_estado

```

```

# A tibble: 54 x 2
  STATE num_aeropertos
  <chr>         <int>
1 TX             24
2 CA             22
3 AK             19
4 FL             17
5 MI             15

```

```

6 NY          14
7 CO          10
8 MN          8
9 MT          8
10 NC         8
# i 44 more rows

```

4

```
library(leaflet)
```

Warning: pacote 'leaflet' foi compilado no R versão 4.5.3

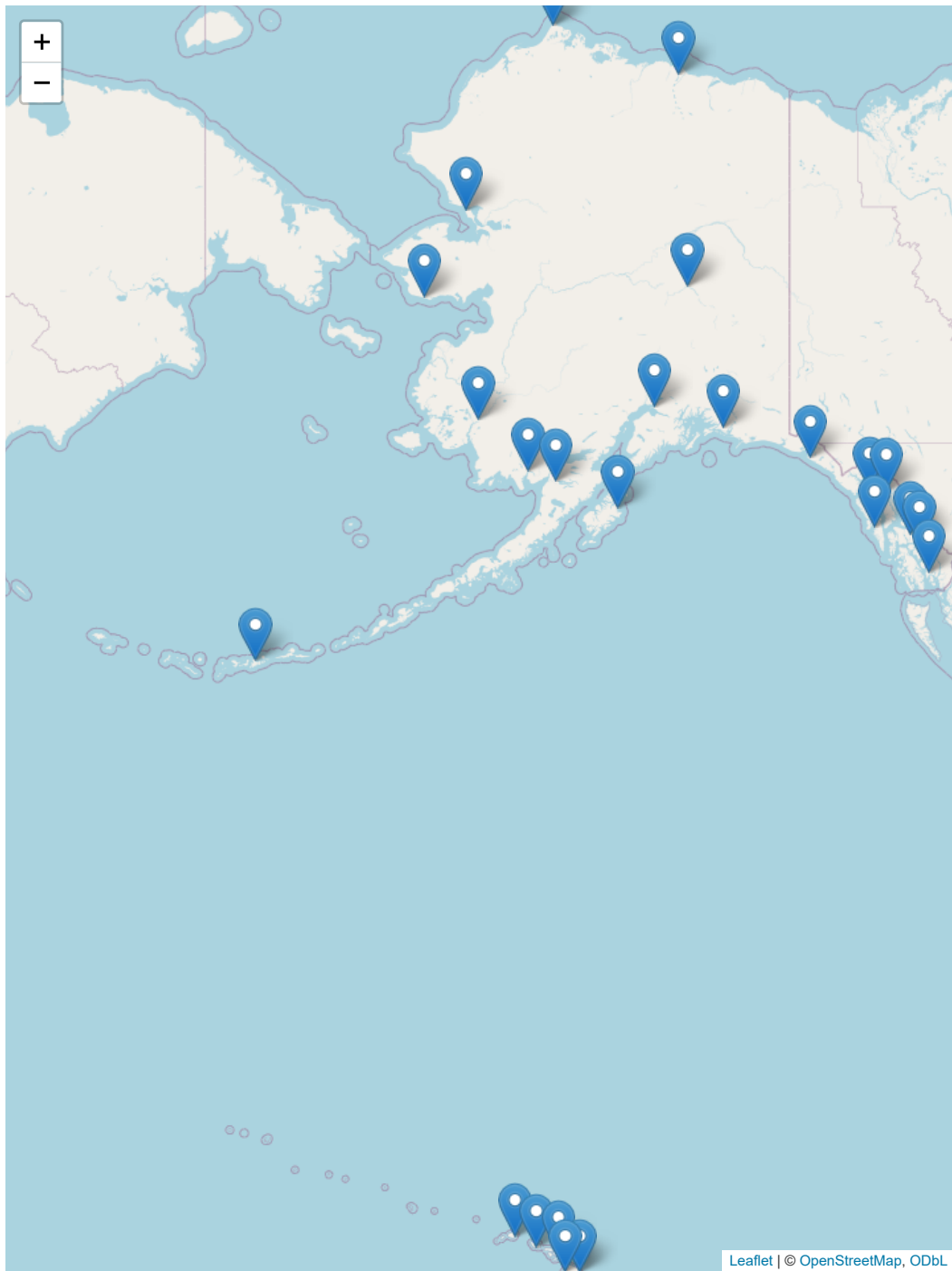
```

flights_mapa <- flights %>%
  select(-any_of(c("CITY", "STATE", "LATITUDE", "LONGITUDE"))) %>%
  left_join(
    airports %>% select(IATA_CODE, CITY, STATE, LATITUDE, LONGITUDE),
    by = c("DESTINATION_AIRPORT" = "IATA_CODE")
  ) %>%
  filter(!is.na(LATITUDE), !is.na(LONGITUDE), !is.na(CITY))

this <- leaflet(flights_mapa) %>%
  addTiles() %>%
  addMarkers(
    lng = ~LONGITUDE,
    lat = ~LATITUDE,
    popup = ~paste0(
      "<b>Aeroporto:</b> ", DESTINATION_AIRPORT, "<br>",
      "<b>Cidade:</b> ", CITY, " - ", STATE, "<br>",
      "<b>Atraso Médio:</b> ", round(atraso_medio, 2), " min"
    )
  )
this

```

Warning in file.info(file): cannot resolve owner of file
 '\\smb\ra266580\Documentos\ME315\file8d44839cc7\webshot8d4a6f6612.pdf': Não foi
 feito mapeamento entre os nomes de conta e as identificações de segurança



```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-08-31 23:10:41 -03"
```